Centre Number	Candidate Number	Candidate Name

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ADVANCED SUBSIDIARY LEVEL 8223/3

PAPER 3 Advanced Practical Skills 2 hours

Marks 40 **2021**

Additional Materials: As listed in the Confidential Instructions to subject teachers.

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the question paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any rough work, diagrams or graphs.
- Do not use correction fluid.
- You may use a non-programmable calculator.
- Do not write in the margin *For Examiner's Use*.

- Answer **all** questions.

- The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

Marker	
Checker	

This document consists of **12** printed pages.

Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

Before you proceed, read carefully through the **whole** of Question 1 and Question 2.

Plan the use of the **two hours** to make sure that you finish the whole of Question 1 and Question 2.

1 The enzyme catalase is found in living cells. Catalase removes toxic hydrogen peroxide (H_2O_2) from cells by breaking it down to form water and oxygen. The equation for this reaction is given below.

$$2H_2O_2 \xrightarrow{\text{catalase}} 2H_2O + O_2$$

You are required to determine how the concentration of hydrogen peroxide affects the rate of catalase enzyme action, by finding the volume of oxygen produced. In this experiment the oxygen produced will be measured by means of the "downward displacement of water" method.

Do the following to correctly set up the equipment:

- Fill the cylinder to the rim with water.
- Place one hand on the open end of the cylinder and turn the cylinder upside down, taking care not to spill any water.
- Place the upside-down cylinder in the trough of water, ensuring that your hand is below the water level of the pan.
- Remove your hand and make sure that the cylinder remains full of water.
- Securely fasten the cylinder in the retort stand (take care that the cylinder remains below the water level and filled with water)
- Place the plastic tubing which is attached to the rubber bung carefully into the cylinder through the open end, as shown in Fig. 1.1

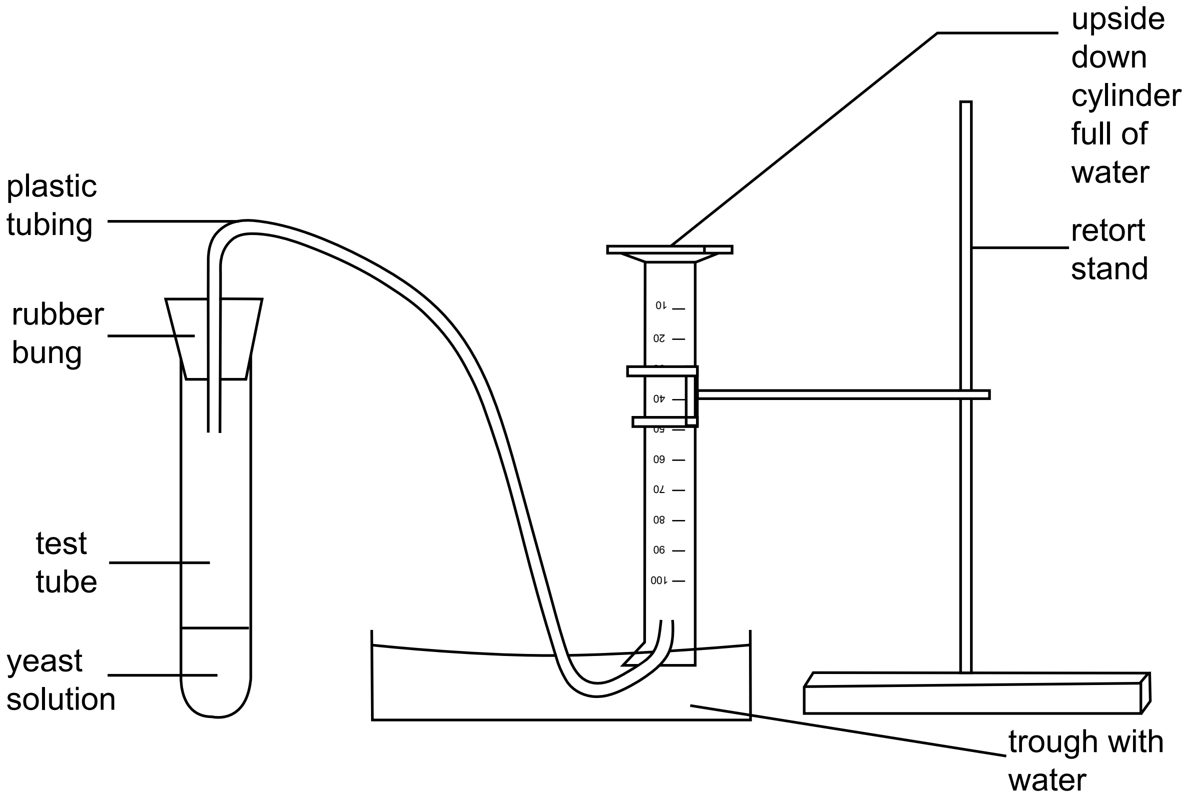

Fig. 1.1

Make sure you wear your rubber gloves when you carry out these procedures.

(a) You are provided with 7 test-tubes labelled **A** to **G**. Each test-tube contains 2 g of yeast cells. Yeast cells contain catalase.

1 Add to each test-tube the volume of water indicated in Table 1.1 and gently shake or stir after having added the water to mix properly.

2 Add to test-tube **A** 4.0 cm^3 of 6% hydrogen peroxide as indicated in Table 1.1. Immediately close the test-tube with the rubber bung that is connected via a plastic tube to the water filled measuring cylinder as shown in Fig. 1.1.

3 As soon as the bung is placed onto the test-tube, start the timer.

4 Gas bubbles formed in the test-tube will push the water out of the cylinder. After 20 seconds pull the plastic tubing from the cylinder to determine how much oxygen has been collected during that time.

5 Record in Table 1.1 the volume of oxygen in cm^3 collected in 20 seconds. [2]

6 Repeat steps 2 to 5 for each of test-tubes **B** to **G** but using different volumes of water and hydrogen peroxide to make up different concentrations of hydrogen peroxide solution shown in Table 1.1.

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Table 1.1

test-tube	volume of water / cm^3	volume of hydrogen peroxide / cm^3	concentration of hydrogen peroxide (%)	time to produce 20 cm^3 of oxygen / s	rate of oxygen production / $cm^3\ s^{-1}$
A	2.8	1.2	6.0		
B	3.0	1.0	5.0		
C	3.2	0.8	4.0		
D	3.3	0.7	3.5		
E	3.4	0.6			
F	3.6	0.4	2.0		
G	4.0	0.0	0.0		

(b) Calculate the hydrogen peroxide concentration for test-tube **E** and record your answer in Table 1.1. [1]

(c) Calculate the rate of oxygen production in $cm^3\ s^{-1}$ for each test-tube and record your answers in Table 1.1 [1]

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(d) (i) Plot a line graph on the grid below to show the effect of hydrogen peroxide concentration on the rate of oxygen production.

[4]

(ii) Use data from the graph to describe and explain the effect of hydrogen peroxide concentration on the rate of catalase enzyme action.

[4]

(iii) Draw a curve on the grid above to show the effect of adding a competitive inhibitor of catalase enzyme action. [1]

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(e) Identify **one** source of possible error in this investigation. Explain the possible effect on this investigation and on the conclusion that can be made.

source of error ....

effect of error ....

....

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....

.... [3]

(f) Catalase enzyme is a protein molecule. A learner decided to test the solution for the presence of this molecule.

Describe how the learner would test and confirm the presence of protein in the solution.

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.... [4]

[20]

2 You are provided with leaf **P** .

You are required to use leaf **P** to perform the following procedure:

1 Paint two strokes of clear nail polish onto the underside of the leaf.

2 Leave the nail polish to dry for approximately 5 minutes.

3 Stick a piece of adhesive tape over the dry nail polish on the underside of the leaf.

4 Slowly peel the adhesive tape from the leaf to remove the nail polish.

5 Stick the adhesive tape attached to the nail polish onto a microscope slide.

6 View the slide at low power using a light microscope.

(a) **(i)** State the overall magnification of the specimen when viewed at low power using the light microscope.

.... [1]

(ii) Make a large drawing of a stomatal pore and six surrounding cells.

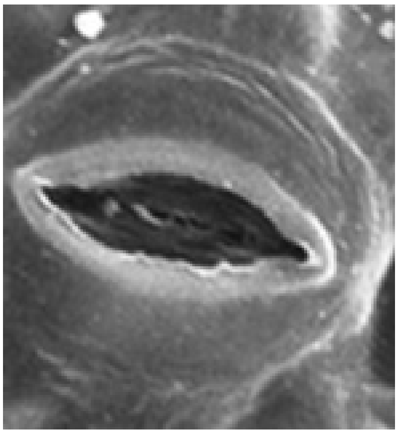
Use ruled label lines to identify two type of cell that you have drawn.

[5]

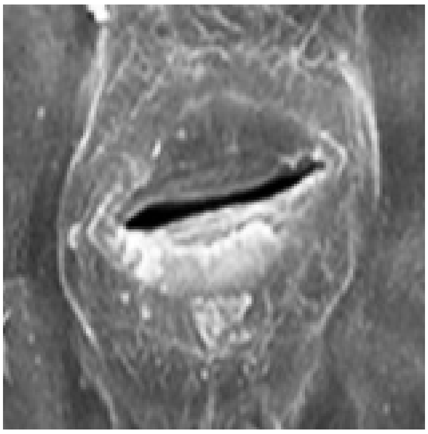
(b) In a similar experiment a learner observed samples of clear nail polish taken from the underside of leaf **Q** and leaf **R**. Each leaf in this second experiment was taken from a separate plant of the same species. Each plant was grown in an environment with a different temperature.

The appearance of part of the nail polish sample from each leaf is shown in Fig. 2.1.

The magnification of both images in Fig. 2.1 is ×800.



leaf **Q**



leaf **R**

Fig. 2.1

(i) Calculate the actual length of the stomatal pore shown for leaf **Q** in Fig. 2.1.

Show your working.

length of stomatal pore = ……..……..…. μm [2]

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(ii) The learner concluded that leaf **Q** was taken from a plant grown in an environment with a higher temperature.

Outline how the learner could modify or extend the investigation to make it a fair test and to ensure the conclusion is reliable.

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.... [3]

(c) Fig. 2.2 shows a cross-section through part of a leaf from a hydrophyte (a plant that lives in water).

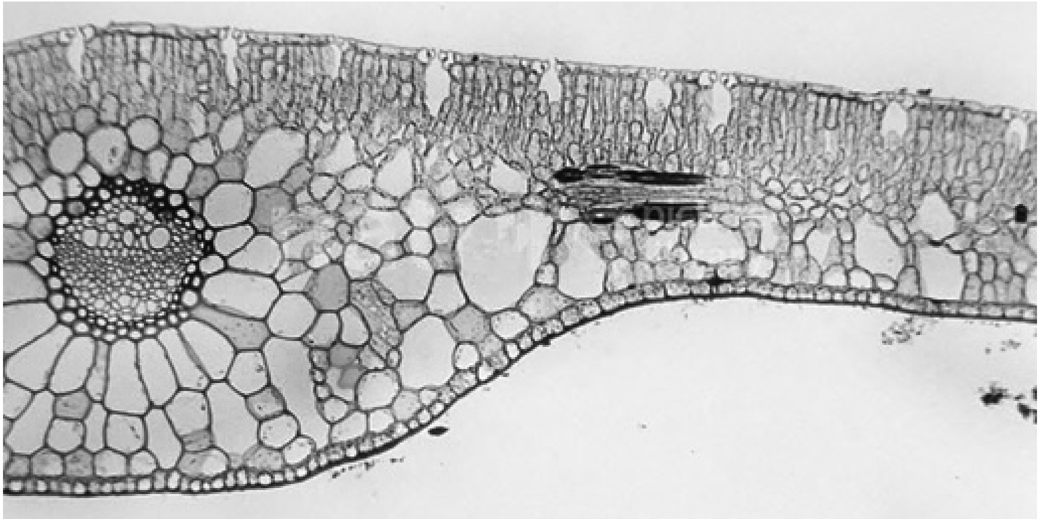

Fig. 2.2

(i) Make a large, labelled plan drawing of the part of the leaf shown in Fig. 2.2.

[4]

(ii) Fig. 2.3 shows a cross-section through part of a leaf from a mesophyte (a plant that requires a moderate amount of water).

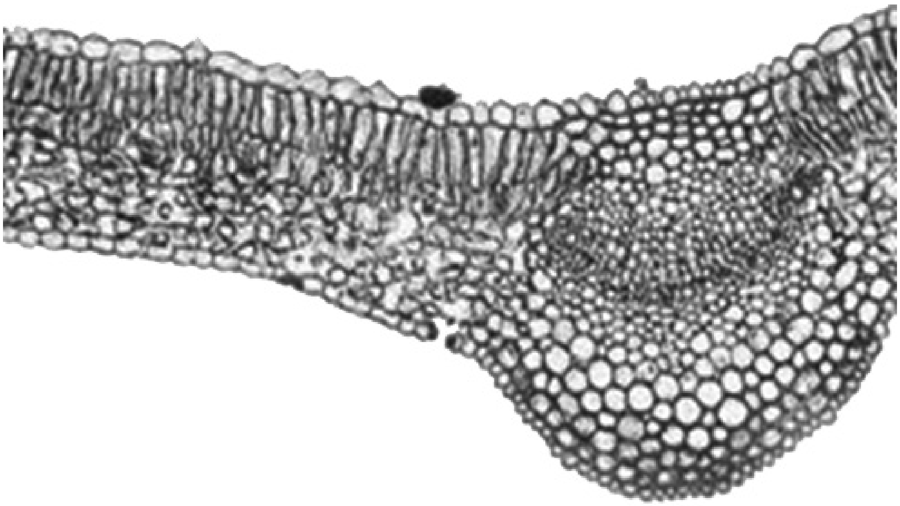

Fig. 2.3

Complete Table 2.1 to compare the named features of the hydrophyte in Fig. 2.2 with the mesophyte in Fig. 2.3.

Table 2.1

feature	**section of leaf from a hydrophyte in Fig. 2.2**	**section of leaf from a mesophyte in Fig. 2.3**
number of stomata		
position of stomata		
size of air spaces		
waxy cuticle		

[4]

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(iii) Describe, with reference to a named feature in Table 2.1, how the leaf of the hydrophyte is adapted to its environment.

....

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.... [3]

[20]